

Final report on the emotion classifier project

Dmytro Dumanskyy Tiago Kienzle Hagen Aleksandar Neshov Daniel Pfefferkorn

{dmytro.dumanskyy, t.hagen, a.neshov, d.pfefferkorn}@campus.lmu.de

Abstract

Facial emotion recognition is a widely researched topic focusing on computationally analyzing and classifying human emotions by visual input. Creating effective AI solutions to automatically and correctly identify a person's sentiment is a technology used in a broad spectrum of sectors. Implementing a neural network capable of fulfilling this requirement is a difficult task that has been the focus of many research papers. The following report shows a valid approach to solving this problem by experimenting with different architectures, training setups and visualisation techniques. The resulting model accurately classifies emotions based on facial cues provided through images

1. Introduction

How do humans communicate?

Most people would immediately think of spoken or written language. Yet communication goes beyond words. An essential component of human interaction is *emotion*. Emotions form a key part of nonverbal communication and have been widely studied within the field of psychology.

1.1. What is facial emotion recognition

Facial emotion recognition (FER) refers to the ability to identify human emotions computationally based on images. Other types of emotion recognition tasks approach the same problem by analyzing speech or writing to reconstruct a person's emotional state. As past research has shown, understanding human emotions is no trivial task for an algorithm. The best results in the field of FER have been achieved using complex model architectures and significant amounts of training data. However, the results are quite promising. Recent publications have exceeded accuracies of 90% [14] in their studies.

1.2. Why is FER important

Given that humans themselves are capable of emotion recognition, it may not be immediately clear why research in this field is so important. The answer to this question is

that characterizing a person's feelings is a key part of many important sectors such as [27]:

- Healthcare
- Public safety
- Crime detection
- Advertising

By automating the emotion recognition process needed in these areas, the risk of human error can be significantly reduced. Furthermore individuals differ substantially in their ability to correctly classify sentiments. Implementing automated emotion recognition solutions therefore provides a consistent baseline for real-world applications.

1.3. Our work

The goal of this project was to research and understand this technology, followed by the construction, training and evaluation of our own AI model specialized in assessing six basic human emotions: happiness, anger, sadness, fear, disgust and surprise.

The following paper will illustrate the approaches we took to optimize our implementation for accuracy.

2. Related Work

2.1. Efficient Architectures and Low-Resolution Vision

The evolution of computer vision architectures has largely been driven by the quest for efficiency and scalability. Tan and Le introduced **EfficientNetV2** [28], highlighting the importance of training speed and parameter efficiency through the use of Fused-MBConv layers. While highly effective on ImageNet, standard CNNs often struggle with information loss when applied to small inputs. Addressing this, Hassani et al. proposed **Compact Convolutional Transformers (CCT)** [12], demonstrating that transformers can escape the “big data paradigm” by using convolutional tokenizers to handle small datasets effectively. Similarly, modern pure ConvNets like **ConvNeXt V2** [33] utilize fully convolutional masked autoencoders to improve representation learning. More recently, **Vision Mamba (Vim)** [35] challenged the transformer dominance by utilizing bidirectional State Space Models (SSMs) for linear

complexity scaling, offering a computationally efficient alternative for visual representation. Our work evaluates these state-of-the-art architectures, adapting their downsampling strategies to the constraints of 64×64 facial expression recognition (FER).

2.2. Transfer Learning in Facial Analysis

Facial Expression Recognition benefits significantly from domain-specific pre-training on massive face recognition datasets to overcome data scarcity. Wang et al. [30] evaluated multiple model architectures and demonstrated the overarching efficacy of transfer learning for establishing robust feature extractors. Specifically, they illustrated that an InsightFace ResNet-50 (IR50) backbone, pre-trained on the MS-Celeb-1M dataset, effectively captures identity-invariant representations prior to processing fine-grained facial expressions. Inspired by this methodology, we adopted the pre-trained IR50 weights from their repository to initialize our network.

2.3. Lightweight Models and Transfer Learning Strategies for Efficient FER

Recent advances in facial expression recognition (FER) have largely been driven by increasingly deep and parameter-heavy convolutional neural networks. However, the CERN [10] architecture challenges this trend by demonstrating that state-of-the-art (SOTA) performance can be achieved with only 1.45 million parameters. Despite its compact design, CERN outperformed several larger and more computationally demanding models across benchmark datasets. On in-the-wild datasets RAFDB and FerPlus CERN reports 87.09

2.4. Optimization and Generalization Techniques

Training deep networks on heterogeneous, imbalanced datasets requires specialized optimization strategies to ensure convergence and generalization.

- **Warm-up and Learning Rate Schedules:** As demonstrated by Goyal et al. [11], utilizing a learning rate warm-up period is critical for stabilizing training, particularly when using large minibatches. This technique prevents the model from diverging due to large gradient updates during the initial epochs when weights are random.
- **Sharpness-Aware Minimization (SAM):** Standard optimizers like AdamW focus solely on minimizing the training loss value. Foret et al. introduced SAM [9], which simultaneously minimizes loss value and loss sharpness. By seeking parameters that lie in “flat minima,” SAM significantly improves generalization performance on out-of-distribution data, a crucial property for our multi-source dataset that includes both lab-controlled and in-the-wild images.

2.5. Choosing a FACS-Aligned Benchmark Dataset for Robust Generalization Testing

CK+ is a well-established dataset focusing on prototypical and micro-expressions, with annotations grounded in the Facial Action Coding System (FACS) developed by Ekman and Friesen [6]. FACS defines facial movements in terms of Action Units (AUs), which correspond to specific underlying muscle activations. The selection of key facial landmarks and their configuration in CK+ follows the FACS framework [20], making it particularly suitable for evaluating models designed to capture fine-grained facial expression dynamics.

3. Methodology

3.1. Datasets

To ensure reliable and comparable results, four widely used benchmark datasets from the literature were selected at the beginning of this study. All datasets contain facial images categorized into the six target expressions: anger, disgust, fear, happiness, sadness, and surprise. The datasets employed were a smaller version of AffectNet [1] (22,629 images), CK+ [3] (927 images), FER+ [8] (56,269 images), and RAF-DB [22] (12,135 images).

FER+ was chosen instead of the original FER2013 due to its improved labeling quality, which has a direct impact on model performance and reliability. Moreover, the selected datasets provide diversity in terms of age, ethnicity, and gender, which is essential for improving the generalization capability of the models. Although AffectNet, CK+, and FER+ include an additional *contempt* label, this class was excluded from the experiments as it falls outside the scope of the present study. The data includes both grayscale and RGB images. Image formats vary between JPEG and PNG, which does not affect the preprocessing or training pipeline.

After the initial training sessions, the developed models exhibited strong overfitting. To mitigate this issue, additional datasets were sought, and access requests were submitted to multiple institutions worldwide to expand the training data.

At the end of the data collection process, a total of 310,215 facial images were gathered. In addition to the previously mentioned datasets, the final corpus also included images from EmoSet [29], JAFFE [16], MMA-FEDB [21], ExpWild [7] and WSEFEP [34]. All datasets were carefully curated and harmonized to match the six target emotion categories used in this study.

Several challenges were encountered during dataset collection and curation. Some datasets contained incorrectly labeled samples, including statues, cartoon characters, or animals assigned to human emotion categories. In other cases, images were organized per subject rather than per

emotion category, without consistent naming conventions, which prevented automated separation. Furthermore, some small datasets were ultimately discarded due to insufficient sample size or poor labeling quality.

As illustrated in the Figure 1, the aggregated datasets are imbalanced, with a significantly higher number of happiness samples compared to the other emotion categories.

3.2. Cross-Dataset Generalization Strategy

For the final training phase, CK+ and KDEF [17] were deliberately excluded from the training set in order to evaluate the generalization capability of the proposed model on unseen datasets. It is important to emphasize that both datasets are relatively balanced.

KDEF contains high-quality, well-controlled facial images captured under consistent lighting and pose conditions, including photos from multiple angles up to 90°. This increased variation in viewpoint makes it more challenging for our models to accurately predict the expressions, which explains their lower performance on KDEF. Although the images were resized to 64×64 pixels to meet the architectural requirements of this study, the dataset remains valuable for assessing cross-dataset robustness due to its standardized acquisition setup.

By excluding CK+ and KDEF from training, these datasets served as independent test sets, enabling a more rigorous evaluation of cross-dataset generalization performance.

3.3. Data Pre-Processing Methodology

The pre-processing pipeline functions as a sequential execution of image evaluation modules, where images failing specific quality thresholds are programmatically discarded. The evolution of this strategy transitioned from strict curation to selective filtering across two empirical phases.

Phase 1: Aggressive Filtering for a Specialized Use Case The primary hypothesis for Phase 1 was that the model would perform optimally if trained exclusively on perfect, high-quality images that mirrored the target demonstration environment (a user looking directly into a high-quality camera). To enforce this pristine standard, the initial pipeline utilized all available filters sequentially:

- **Lighting Filter:** Analyzed the mean pixel intensity and standard deviation, discarding images that were excessively dark, overly bright, or lacked sufficient contrast.
- **Blur Detection:** Assessed image sharpness using frequency analysis via a Fast Fourier Transform (FFT). Images lacking sufficient high-frequency components were rejected.
- **Roll Correction:** Evaluated the image across four rotational states (0° , 90° , 180° , 270°), selecting the orientation that yielded the highest face detection confidence.

- **Yaw Filtering:** RetinaFace detected the face, and 6DRepNet predicted the 3D pose. To guarantee a strict frontal viewpoint, any image with a yaw angle exceeding 45° was discarded.
- **Conditional Cropping:** If the ratio of the face area to the total image area was less than 0.4, indicating a small face with significant background, the image was cropped with proportional padding; otherwise, the original image was retained.

In Phase 1, we retained approximately 91,960 images after aggressive filtering.

Phase 2: Resolving the Augmentation Contradiction

The aggressively filtered dataset from Phase 1 caused the models to overfit rapidly. To combat this, data augmentation techniques were applied (adding artificial noise, altering brightness, introducing artificial blur, and utilizing MixUp).

This sequence exposed a fundamental contradiction: we were actively removing naturally difficult images during the pre-processing stage, only to artificially recreate those exact conditions during the augmentation stage. Furthermore, augmentations like MixUp produce overlapping visual structures that standard face detectors completely fail to recognize, meaning such images would be universally rejected by strict filters. Recognizing that the model required this data variance to generalize, Phase 2 drastically reduced the aggression of the filtering pipeline:

- **Reduced Aggression:** The universal application of blur, yaw, roll, and lighting filters was halted. For high-quality datasets (AffectNet, CKplus, FERPlus, RAF-DB), the pipeline was stripped down to utilize only structural standardization.
- **Selective Cleaning:** Aggressive techniques were retained only for uncleaned, chaotic datasets (e.g., ExpW and EmoSet), which utilized the cropping module, face detection validation, and manual exclusion of specific emotion labels.

In Phase 2, we started with 310,215 images. After selective pre-processing, we retained 229,478 images.

Standardization and Deduplication To finalize the datasets, all images underwent structural standardization:

- **Channel Normalization:** All inputs were converted to a 3-channel RGB format to ensure consistent tensor depth.
- **Resizing:** Images were zero-padded to achieve a square aspect ratio and subsequently resized to exactly 64×64 pixels. We utilized cubic interpolation if the original image was smaller than 64×64 .
- **Deduplication:** Duplicate or highly similar images were purged using a combination of perceptual hashing (pHash) and difference hashing (dHash) to ensure strict separation between training and validation distributions.

Dataset Splitting At the beginning, we followed an 80%/10%/10% split strategy, where 80% of the data was

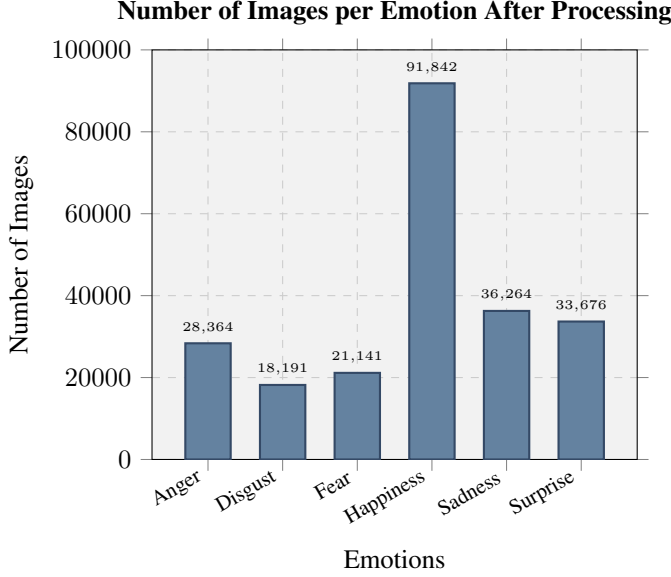


Figure 1. Number of Images per Emotion After Processing

used for training, 10% for validation, and 10% for testing. Towards the end of the project, some of us adopted a 90%/10% split strategy in most experiments. The training set comprised 90% of the data, while 10% was reserved for validation (evaluation). For the final test set, we exclusively utilized two specific high-quality datasets: CK+ and KDEF. This rigorous separation ensured that our final performance metrics reflected true generalization on standardized benchmarks.

3.4. Architectural Adaptations

3.4.1 Basic CNNs and ResNet18 models

To ensure experimental comparability, two baseline convolutional neural networks (3-layer and 5-layer CNNs) were implemented alongside a modified version of ResNet-18, in which the initial 7×7 convolution was replaced with a 3×3 kernel and the input resolution was reduced from 224×224 to 64×64 .

3.4.2 ResNet18 with Squeeze-and-Excitation

Our model is based on ResNet18 and is extended with a squeeze-and-excitation (SE) attention module. ResNet18 serves as a stable baseline with moderate capacity (roughly 11–12 million parameters). Results from compact FER literature, including CERN, suggest that attention mechanisms can improve performance across different model sizes, supporting our use of SE as a low-overhead architectural enhancement [10].

The input to our network is a standardized RGB face image of size 64×64 . Since this is much smaller than the

standard ImageNet setting (224×224), we avoid the original ResNet stem (7×7 stride-2 convolution plus stride-2 max pooling), because it would reduce spatial resolution too early and remove fine facial details. Instead, we use a 3×3 convolution with stride 1 and omit early pooling to preserve spatial information [25].

After the stem, we follow the standard ResNet18 design with four residual stages and stage-wise downsampling. Skip connections stabilize optimization by learning residual refinements rather than fully rewriting features [13]. After the final stage, global average pooling and a dropout-regularized linear layer produce logits for the six emotion classes.

We augment the ResNet18 backbone with squeeze-and-excitation (SE) blocks, which perform lightweight channel-wise attention by learning per-channel scaling factors from globally pooled features, improving representational capacity with minimal overhead [15]. The motivation for using a lightweight attention mechanism is consistent with the goal of improving robustness without greatly increasing model size and is aligned with compact FER approaches such as CERN [10].

3.5. Class Imbalance and Advanced Loss Functions

A significant challenge encountered in our training pipeline is the severe class imbalance within our aggregated dataset, where the *happiness* class constitutes approximately 40% of the total data. Under standard cross-entropy loss, this imbalance introduces a fundamental bias where the majority class becomes the default, high-confidence prediction. This phenomenon is driven by two factors:

- **Frequency versus Difficulty:** The *happiness* class typically presents distinct visual markers (e.g., a smile) that are easily learned. Conversely, minority classes such as *fear* and *disgust* rely on subtle ocular and brow micro-expressions that degrade heavily at lower resolutions, such as 64×64 pixels.
- **Loss Gradients:** Because majority classes occur more frequently, they contribute a disproportionately large share of the gradient updates. The model minimizes the global objective simply by maximizing confidence in the frequent, easy classes, functionally neglecting the minority classes.

To address this imbalance without relying on a `WeightedRandomSampler`, we modified the training objective by evaluating two advanced loss functions:

- **Class-Balanced Focal Loss (CB-FL):** This loss was applied to **both the ResNet18+SE and EfficientNetV2-S models**. It assigns larger weights to underrepresented classes using the effective number of samples [5], while the focal component reduces the contribution of well-classified (easy) samples so that the model continues learning from harder examples [19].

- **Label-Distribution-Aware Margin Loss (LDAM) with Deferred Re-Weighting (DRW):** This loss was evaluated **exclusively on the EfficientNetV2-S model**. LDAM modifies the logit outputs by assigning a significantly larger mathematical margin to minority classes [2]. This strictly penalizes the model unless it maintains extreme confidence when predicting rare classes, thereby enforcing wider decision boundaries around them. The DRW strategy schedules the training such that the model trains without artificial weights for the initial epochs to learn robust feature representations. In the final epochs, strict class frequency weights are applied to fine-tune the decision boundaries for the minority classes without corrupting the initial feature learning phase.

3.5.1 Transfer Learning (IR50)

We wanted to try bigger models for our project. However, we did not have enough data to train a large model from zero. Transfer learning allowed us to test how this method works in our specific case and opened the possibility to use a larger architecture. A positive side effect was that the pre-trained model (InsightFace ResNet-50) had already learned facial features from a massive dataset of one million images (Ms-Celeb-1M) [30].

The adaptation occurred in three steps:

Version 1: Original Full Fine-Tuning The first test used the entire IR50 architecture, including the original classification head.

- **The Problem:** The original head flattened the spatial grid into a 25,088-dimensional vector and sent it into a massive linear layer. This single layer had 12.8 million random parameters, accounting for almost half of the whole model.
- **Result:** Training all 27 million parameters on small datasets caused rapid overfitting. The massive linear layer memorized exact spatial positions, and the high learning rate ruined the pre-trained weights.

Version 2: Layer Freezing and the Memorization Problem To protect the pre-trained feature extractors, we froze the early layers (`input_layer`, `body1`, `body2`, and `body3`), which contain about 9 million parameters.

- **Methodology:** We could not freeze the entire backbone because emotion classification requires the network to identify specific facial features that a face recognition model might ignore. Therefore, we kept the final convolutional block (`body4`) trainable.

Version 3: The Light Head Architecture To stop the memorization while keeping `body4` trainable, we built a new head for the model.

- **Methodology:** We removed the 12.8-million-parameter head and replaced it with a “Light Head” containing only 795,000 parameters.

- **Spatial Attention Integration:** Using a standard Global Average Pooling (GAP) layer fixes the parameter problem but destroys spatial information. Human emotions are local, making spatial context critical. We added a Spatial Attention module (Conv1x1 followed by a Sigmoid activation) before the GAP layer. This module learns to give weights to important spatial areas before the GAP layer reduces the dimensions.

3.5.2 Alternative Architectures

During the evaluation phase, three low-parameter architectures were tested to compare different modern deep learning paradigms. Adapting these models for 64×64 inputs required specific architectural modifications, primarily reducing spatial downsampling.

- **Vim (Vision Mamba, ~7M parameters):** Vim [35] was chosen because it is a novel model challenging standard Transformers by utilizing bidirectional State Space Models (SSMs), promising exceptionally low GPU memory usage. To adapt it for 64×64 images, the patch size was changed from 16×16 to 8×8 to provide a functional sequence length of 64 tokens.
- **CCT-7 (Compact Convolutional Transformer, ~4M parameters):** CCT-7 [12] was selected to test a small Transformer specifically designed to handle small images by using a convolutional tokenizer instead of linear patch embeddings.
- **ConvNeXt V2 Atto (~3.7M parameters):** This model [33] was chosen to evaluate how a state-of-the-art pure Convolutional Neural Network operates in a constrained parameter regime. The original architecture features a stem convolution with a stride of 4. For a 64×64 image, this immediately shrinks the feature map to 16×16 . The stem stride was explicitly changed to 1 to preserve spatial information.

3.5.3 EfficientNetV2-S Adaptations

EfficientNetV2-S [28] was selected for its MBConv blocks and Squeeze-and-Excitation (SE) attention, which effectively amplify emotional markers like micro-expressions while suppressing background noise.

Architectural and Augmentation Adaptations To process 64×64 images effectively, two major pipeline adjustments were executed:

- **Stem Stride Fix:** The original stem stride of 2 was reduced to 1. This modification preserved the full spatial resolution through the initial layers, preventing the immediate loss of fine facial details.
- **Minimized Destructive Augmentation:** Initial training with standard ImageNet settings (Mixup $\alpha = 0.8$) obliterated facial features. We implemented a low-interference pipeline to minimize data destruction while maintaining

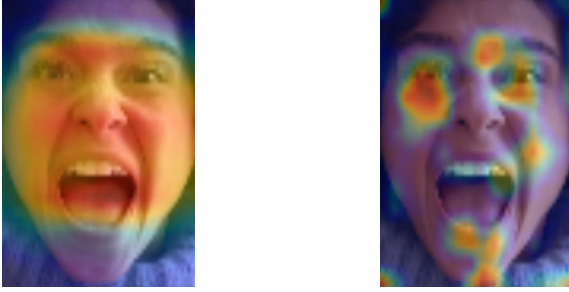


Figure 2. Grad-CAM result of the fourth convolutional layer (left) and the third convolutional layer (right)

regularization. This included reducing Mixup to 0.2, limiting CutMix to a maximum of 0.3, and using gentle affine transformations. This ensured that while the model was forced to reconstruct features, the core emotional signal remained intact.

3.6. Explainable AI

AI systems that only expose the input and output of the model to the user are called "black box models". The difficulty this design poses lies in the fact that the developer cannot clearly understand the model's reasoning process. Explainable AI aims to solve this problem by providing a method to visualize the importance of different image regions to the final decision of the model. For our task we experimented with two different methods:

3.6.1 Grad-CAM

The grad-cam method is widely used to explain computer vision models. The algorithm works by analyzing the gradients used by the last convolutional layer and assigning a level of importance for each neuron's output for the final classification of the image [23]. However this approach did not seem to work as well as we hoped for our model. Applying grad-cam to our model's last convolutional layer resulted in a coarse and inaccurate heatmap. To combat this problem we experimented with using the previous convolutional layers as the base layer for grad-cam which yielded strongly improved results as in Figure 2.

The poor performance of the grad-cam method is a result of the heavy downsampling during our model's forward pass. The resolution (8×8) of the final feature maps lacks sufficient spacial information to effectively reconstruct the importance of the low resolution input. This also explains why we achieved better results on a layer using higher resolution feature maps.

3.6.2 Integrated gradients

Integrated gradients is a different algorithm to assess the importance of different image regions for the model's decision. This method works by attributing importance values to the pixels of a baseline image. This image is slowly transformed into the original input image and in each transformation step the gradients of each pixel are calculated. The Importance of each pixel is then set as the average gradient of each pixel, which represents how much the model's output changes based on that piece of information [26]. The result is a fine grained attention map which highlights the most important parts of the image as seen in Figure 3.

This approach seemed interesting to us as the quality of the output did not rely on the architectural details of our model.

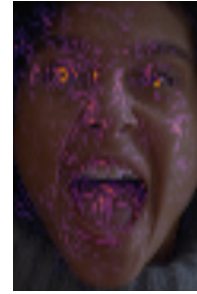


Figure 3. Heatmap resulting from the integrated gradients method

4. Experiments and Results

4.1. Basic CNNs and ResNet18 Configuration Enhancements

The performance of ReLU and LeakyReLU activation functions was systematically evaluated, with LeakyReLU demonstrating superior overall accuracy. Optimization strategies included Stochastic Gradient Descent (with standard hyperparameters), Adam (with learning rate set as 0.001), and AdamW were tested and Adam with a learning rate of 0.001 yielded the best performance across both the shallow CNN architectures and the modified ResNet-18. Further improvements were achieved through the integration of Batch Normalization, Dropout regularization, and late-stage global average pooling, which enhanced feature stability, mitigated overfitting, and preserved high-level spatial representations relevant to emotion classification.

4.2. ResNet18+SE Ablation: Optimizer and Loss Function

To measure the effect of optimizer choice and loss function while keeping the architecture fixed, we trained the same ResNet18+SE model under four configurations: (1) AdamW + class-balanced focal loss, (2) AdamW +

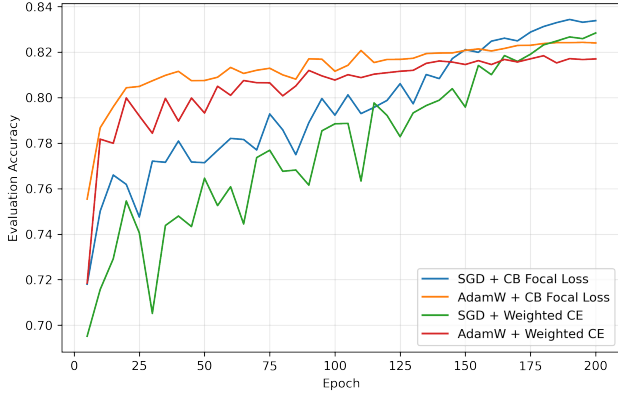


Figure 4. Validation performance comparison for ResNet18+SE under different optimizer and loss configurations.

weighted cross-entropy, (3) SGD + class-balanced focal loss, and (4) SGD + weighted cross-entropy. All runs used the same input resolution (64×64), the same augmentation setup, and the same training duration. The validation performance curves are shown in Fig. 4.

Across the four settings, SGD with class-balanced focal loss achieved the strongest validation performance. One likely reason is that AdamW can converge quickly early on, but with noisy and heterogeneous FER data it can also lead to solutions that generalize less well. In addition, class-balanced focal loss tends to be more robust for strong imbalance than weighted cross-entropy because it continues to emphasize hard and minority-class examples later in training [4, 18]. In the final pipeline, imbalance is therefore handled mainly through the loss formulation rather than through sampling-based balancing.

4.3. Transfer Learning (IR50)

Version 1: Original Full Fine-Tuning The first test used the entire IR50 architecture, including the original classification head.

- **The Problem:** The original head flattened the spatial grid into a 25,088-dimensional vector and sent it into a massive linear layer. This single layer had 12.8 million random parameters, accounting for almost half of the whole model.
- **Result:** Training all 27 million parameters on small datasets caused rapid overfitting. The massive linear layer memorized exact spatial positions, and the high learning rate ruined the pre-trained weights.

Version 2: Layer Freezing and the Memorization Problem

- **Result:** We applied different learning rates, but the network still possessed the oversized 12.8-million-parameter head. This setup led to severe overfitting, shown by a performance gap of over 10 percent between training and

validation metrics. The massive parameter count allowed the model to memorize the training data.

Version 3: The Light Head Architecture

- **Result:** With only 5.8 million trainable parameters, a reduced MixUp probability, and an Exponential Moving Average (EMA) decay, this setup successfully stopped the overfitting gap and achieved a peak validation accuracy of 81.38 percent.

4.4. Alternative Architectures Evaluation

4.4.1 ResNet18+SE vs. ResNet18+CBAM

To evaluate whether a stronger attention mechanism improves cross-dataset generalization, we replaced the squeeze-and-excitation (SE) blocks in our ResNet18 backbone with CBAM [32] and compared both variants under the same training protocol. In both cases, the models were trained on the aggregated training corpus while explicitly excluding CK+ and KDEF. These two datasets were then used as independent test sets to measure generalization to unseen, well-controlled benchmark distributions.

Overall, both attention variants performed similarly in the sense that they achieved reasonable accuracy on the held-out datasets, but SE consistently yielded higher test accuracy. ResNet18+SE reached 84.69% on CK+ and 67.33% on KDEF, whereas ResNet18+CBAM achieved 81.97% on CK+ and 62.67% on KDEF. On the global held-out evaluation set ("All"), ResNet18+SE achieved 68.47% compared to 63.94% for ResNet18+CBAM. These results suggest that the additional spatial attention component in CBAM did not translate into improved robustness for our low-resolution (64×64) FER setting, while the simpler channel-only SE mechanism provided slightly more stable generalization across datasets.

4.4.2 Vim (Vision Mamba):

However, Vim is fundamentally designed for high-resolution data, and it proved too difficult to make the architecture focus effectively on the fine details of small facial images.

4.4.3 CCT-7 (Compact Convolutional Transformer):

Even though the architecture was perfectly suited for small images, it simply possessed too few parameters to learn the complex features required for human emotion classification.

4.4.4 ConvNeXt V2 Atto:

Similar to CCT-7, the model lacked the parameter count necessary to build robust representations.

4.4.5 MobileNet V2

With $\approx 3.5\text{M}$ parameters MobileNet V2 is one of the smallest Networks we implemented. Sadly the low parameter count and adaptation to images of size 64×64 were incompatible with our goal of effective emotion recognition. The model only achieved an accuracy of 66% on the evaluation data and was hence discarded from future experiments.

4.4.6 Other ResNet implementations

Regarding the high accuracy of the ResNet18 + SE implementation ResNet34 and ResNet50 were also trained and evaluated. However the training process was less efficient since the models started overfitting early on due to the high parameter count. As a result the model evaluation yielded a lower accuracy than the original ResNet18 + SE design.

4.5. EfficientNetV2-S Optimization Dynamics

EfficientNetV2-S ultimately emerged as one of the top two best-performing models across all validation benchmarks. To ensure a rigorous comparison, we empirically optimized and tuned each optimizer to find its stable configuration.

Experiment 1: SGD with Nesterov Momentum

- **Result:** Peaked at 74.25 percent validation accuracy.
- **Performance:** SGD [11] proved highly sensitive to hyperparameters. Slight alterations in the learning rate caused massive performance swings, ranging from a complete failure to learn (stagnating below 40 percent accuracy) to achieving a moderate peak of approximately 75 percent.
- **Why it underperformed:** Even at optimal settings, SGD struggled with structural incompatibilities. It applies a uniform learning rate across the network, causing it to under-train depthwise layers while over-training pointwise layers. It lacks the variance tracking required to stabilize the contradictory signals arising from 10 distinct data sources, and it struggles to converge when targets are ambiguous soft labels from Mixup.

Experiment 2: AdamW

- **Result:** Peaked at 83.10 percent validation accuracy.
- **Why it succeeded:** AdamW's adaptive per-parameter learning rates successfully navigated the heterogeneous architecture, balancing updates between depthwise and dense layers. It provided consistent convergence but exhibited mild overfitting in the final epochs.

Experiment 3: SAM + AdamW (Sharpness-Aware Minimization)

- **Result:** Peaked at 82.46 percent validation accuracy.
- **Performance Dynamics:** SAM [9] improves generalization by minimizing the worst-case loss in the immediate neighborhood of the parameters, forcing the model into flat, stable minima. While it slightly underperformed

the standard AdamW on the validation dataset, it demonstrated superior generalization stability. The drop in accuracy between the validation and test datasets was smaller for SAM, indicating robust out-of-distribution performance. However, its final absolute accuracy on the test dataset remained nearly identical to the simple AdamW baseline, differing by approximately -1 percent. Ultimately, this marginal improvement in generalization stability was outweighed by the severe computational burden of evaluating the network twice per step.

4.6. Evaluation of Class Imbalance and Prediction Bias

Evaluating the EfficientNetV2-S baseline model exposed a severe prediction bias: despite a global accuracy of 60.17%, the network over-predicted *happiness* (96.62%) while failing on *fear* (17.20%) and *disgust* (29.99%). To address this disparity, we evaluated two advanced anti-bias functions: Class-Balanced Focal Loss (CB-FL) and Label-Distribution-Aware Margin Loss (LDAM) with Deferred Re-Weighting (DRW).

While these techniques successfully forced the network to fixate on difficult images, performance improvements were strictly limited to the training dataset. During true cross-dataset testing on unseen CK+ and KDEF faces, minority class accuracies remained stagnant or declined, and global accuracy fell by approximately 3%. This performance degradation stems from three primary factors:

- **Dataset Memorization:** Chasing strict mathematical fairness on small, noisy distributions caused the models to overfit to specific training lighting and identities rather than generalizable features.
- **Data Ambiguity:** Subjective emotions like *fear* and *disgust* contain inherent label noise. Forcing the algorithm to perfectly separate these overlapping concepts exacerbates overfitting.
- **Hyperparameter Tuning:** The degradation might stem from suboptimal configurations. For example, setting margin boundaries or focal penalties too aggressively can severely disrupt generalization, indicating a need for further systematic tuning to discover the optimal balance.

4.7. GAN

Inspired by the work with Super-Resolution Generative Adversarial Network [24] we decided to test with a 2x upscaling factor applied at our 64x64 images and then resize it back to 64x64.

Furthermore, a Super Resolution Generative Adversarial Network (SRGAN) method is a GAN that can convert low-resolution images into high-resolution and realistic images. [31]

Unfortunately, due to the image size and that the enhancement process makes the images a bit softer with less

facial marks, the training with GAN-enhanced images was worse than with the normal images, achieving easily more than 15% difference between train and evaluation accuracy.

The chosen GAN was GFPGAN, a practical algorithm for real-world face restoration. The model with the second version had the best experimental result and was selected to be applied to all the images, so the group could compare between the normal folder and the GAN-enhanced folder.

4.8. Dual-Input Training

Inspired by the paper about CERN [10] the dual-input strategy was implemented. It consists of giving the training process the same image as two inputs and the correct label. The first image is the normal one and the second is augmented. So in order to receive the correct result, the model has to predict both versions of the image correctly. The results with the dual-input were outperformed by the normal training. It also has increased overfitting, a difference bigger than 10% (86.45% to 76.15%) between train and evaluation accuracy.

4.9. CERN-Based Architecture

The CERN architecture is derived from a LightCNN backbone with depths ranging from 9 to 29 layers in its full configuration. It employs Max-Feature-Map (MFM) activations for competitive feature selection and integrates Efficient Channel Attention (ECA) to perform lightweight channel-wise recalibration. A central component of the framework is its dual-input training strategy, where the network processes both an original image and its augmented counterpart simultaneously. Correct prediction requires label consistency across both inputs, promoting robustness and invariance to perturbations.

Due to GPU memory constraints, it was not feasible to replicate the complete architecture with combined local and global attention mechanisms. Therefore, a reduced variant consisting of seven convolutional layers and a batch size of 16 was implemented to fit within the available computational resources. Even under this simplified configuration, training remained computationally intensive: 10 epochs required more than 24 hours, and long runs were occasionally interrupted, necessitating continuation from saved checkpoints.

At epoch 47, the model achieved 66.70% evaluation accuracy and 67.51% training accuracy. The learning dynamics showed extended plateaus followed by sudden improvements, suggesting that additional training time could potentially yield further performance gains. However, limited computational capacity—including the use of an NVIDIA RTX 2060 GPU—prevented full exploration of longer training schedules or the use of pretrained weights from the original seven-class model.

5. Conclusion

This project successfully developed a comprehensive emotion classification system for facial expressions. We implemented a complete pipeline from raw image data through preprocessing to model training and evaluation, utilizing state-of-the-art deep learning techniques.

5.1. Key Contributions

1. **Robust Preprocessing Pipeline:** Developed comprehensive preprocessing procedures including face detection, alignment, deduplication, and quality filtering that significantly improved dataset quality and model robustness.
2. **Attention-Enhanced Architecture:** Integrated Squeeze-and-Excitation blocks into ResNet-18, demonstrating consistent performance improvements over baseline models.
3. **Class-Balanced Training:** Employed focal loss to address class imbalance, resulting in more balanced performance across emotion categories.
4. **Multi-Dataset Evaluation:** Comprehensive evaluation across multiple emotion datasets demonstrated model generalization capabilities and robustness across domain shifts.
5. **Model Interpretability:** Applied Grad-CAM visualization to validate that the model learns emotion-specific facial features, providing insight into the decision-making process.

5.2. Limitations and Future Work

While our system achieves competitive performance, several limitations and opportunities for future work exist:

5.2.1 Limitations

- Performance degradation on low-quality images and extreme face angles
- Difficulty distinguishing between similar emotions (fear/surprise, anger/disgust)
- Limited robustness to occlusions and non-frontal faces
- Dependency on extensive preprocessing and high-quality face detection

5.2.2 Future Directions

- Exploration of transformer-based architectures (Vision Transformers, Compact Transformers) for potentially improved performance
- Integration of temporal information for video-based emotion recognition
- Development of robust multi-task learning approaches combining emotion, age, gender, and identity recognition

Model Accuracy vs. Parameter Count

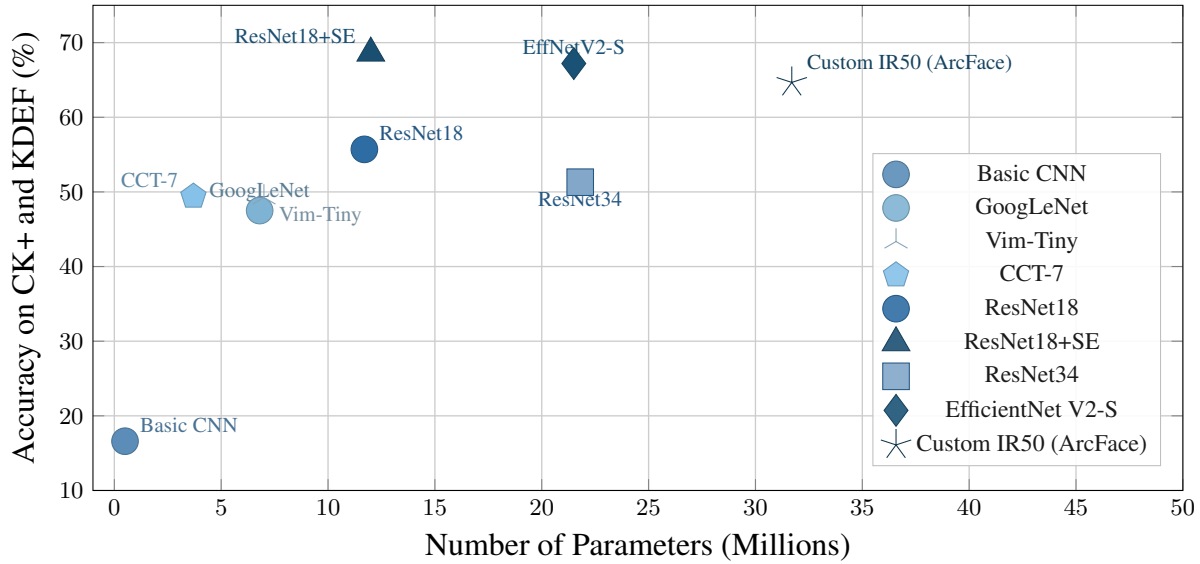


Figure 5. Model Accuracy vs. Parameter Count evaluated on the CK+ and KDEF datasets.

- Investigation of domain adaptation techniques to improve cross-dataset generalization
- Incorporation of action unit detection for finer-grained emotion understanding
- Real-time deployment optimization for mobile and edge devices

5.3. Final Remarks

The facial emotion classification system presented in this report demonstrates the effectiveness of combining careful data preprocessing, attention mechanisms, and appropriate loss functions for emotion recognition tasks. The model's interpretability through Grad-CAM analysis confirms that it learns meaningful emotional representations. Future work should focus on improving robustness to challenging conditions and exploring more advanced architecture designs, particularly with emerging transformer-based models. The comprehensive multi-dataset evaluation framework established in this project provides a solid foundation for continued research in this important area of computer vision.

A. Statement of Contributions

Dmytro

Model Training and Experimentation: Led the training and evaluation of EfficientNetV2-S (and its associated configurations), CCT-7, Vim-tiny, and explored the ConvNeXt V2 architecture. Conducted transfer learning experiments utilizing the IR50 backbone. Executed the final robustness tests for the ResNet18+SE

model across various hyperparameters, which confirmed baseline stability despite yielding identical final accuracies.

Data Pipeline and Preprocessing: Enhanced and developed segments of the data preprocessing pipeline by implementing Fast Fourier Transform (FFT) for an improved version of blur filtering, introducing facial roll correction, yaw correction, and optimizing boundary margin resizing to ensure minimal residual margins post-cropping or square image outputs. Collaborated directly with Tiago to determine and execute the optimal combination of preprocessing operations tailored to each specific dataset.

System Integration and Explainability: Engineered a live demonstration script designed to process video input and output continuously annotated video streams, integrating real-time emotion classification alongside Explainable AI (XAI) visual overlays.

Research and Organization: Sourced approximately 90 percent of the datasets introduced during the second project phase. Conducted extensive literature reviews to identify and propose alternative network architectures. Facilitated project management by structuring team meetings and designed the foundational architecture of the project's README documentation.

A. Extended Description of the CERN Architecture Implementation

The CERN architecture is derived from a LightCNN backbone with depths ranging from 9 to 29 layers in its full configuration. LightCNN is designed to provide compact yet discriminative feature representations, particularly for face-related tasks. It employs Max-Feature-Map (MFM) activations, which perform competitive feature selection by suppressing weaker feature responses through channel-wise max operations. This mechanism reduces redundancy, improves generalization, and lowers computational overhead compared to conventional activation functions.

To enhance representational capacity without significantly increasing complexity, the architecture integrates Efficient Channel Attention (ECA). Unlike heavier attention mechanisms, ECA performs lightweight channel-wise recalibration using local cross-channel interaction without dimensionality reduction. This allows the model to emphasize informative feature channels while maintaining computational efficiency.

A central component of the framework is its dual-input training strategy. During training, the network processes both an original image and its augmented counterpart simultaneously. The prediction is considered correct only if label consistency is maintained across both inputs. This constraint encourages the network to learn perturbation-invariant and robust feature representations, improving generalization under variations such as illumination changes, noise, and geometric transformations.

Due to GPU memory constraints on the university server, it was not feasible to replicate the complete architecture including the combined local and global attention modules described in the original formulation. Therefore, a reduced variant consisting of seven convolution layers was implemented. The batch size was limited to 16 samples to remain within available memory limits.

Despite these simplifications, the model remained computationally demanding. Training for 10 epochs required more than 24 hours of runtime. Extended training sessions were occasionally interrupted due to system constraints, making it necessary to implement checkpoint-based continuation to preserve training progress and ensure experimental reproducibility.

B. Contributions

Dmytro

[TODO: Add contrib]

Aleksandar

[TODO: Add contrib]

Tiago

- Conducted literature research on CNNs, ResNet architectures, GANs, data augmentation techniques, and face detection methods to define the project's technical direction.
- Performed dataset investigation and detailed quality analysis, including data distribution, class balance, image consistency, and label reliability. Compared original datasets with GAN-augmented datasets to evaluate performance impact.
- Implemented baseline CNN models with 3 and 5 convolution layers and adapted ResNet-18 to the project scope. Conducted extensive experiments involving batch normalization, variations in average pooling, dropout tuning, data augmentation strategies, dual-input training setups, and comparative evaluation using GAN-generated images.
- Adapted and optimized the GAN implementation to align with project requirements and ensure compatibility with the university server environment.
- Developed preprocessing pipelines including resizing and cropping using MTCNN, later replacing it with RetinaFace, and analyzed their impact on model performance.

Daniel

Duplicate finder : A class that finds duplicate images

This class can be configured the pHash or dHash algorithm for fuzzy hashing. It exposes a method called `find.duplicates` expects a list of filepaths as an input. The given files are then loaded in batches, hashed and stored in a Dictionary if the hamming distance is higher than a given threshold. The Dictionary is used to avoid unnecessary comparisons (images are in the same "Bucket if the first 32 Bits match"). The method returns a list of paths to duplicate images for further processing.

Blur filter : A class that measures the blurriness of a given image

This class is part of the group's preprocessing pipeline. Given an image the class computes the laplacian variance of the input and removes it if the variance is lower than a given threshold. This implementation was later adapted to use the *fast fourier transform* (FFT) to classify the image.

Different Model implementations : Different custom implementations that were trained

During the development phase of the project I wanted to evaluate as many different model architectures as possible. Due to the fixed input size of (64×64) all models were implemented from scratch and modified to downsample less aggressively. The developed models were all trained using a custom training script.

Model Evaluation script : A script that given a Model implementation measures its accuracy

This script was later used by all members of the group to measure and compare their model's performance on unseen datasets.

Image demo : A script that processes all images in a directory

This script expects the path to a directory filled with images, loads all images in batches, loads our model and evaluates all images. The resulting probabilities for each emotion are then written into a csv file with the related filepath

References

- [1] AffectNet. Affectnet dataset. <https://www.kaggle.com/datasets/mstjebashazida/affectnet>.
- [2] Kaidi Cao, Colin Wei, Adrien Gaidon, Nikos Arachis, and Tengyu Ma. Learning imbalanced datasets with label-distribution-aware margin loss. In *Advances in Neural Information Processing Systems*, 2019.
- [3] Cohn-Kanade. Ck+ dataset. <https://www.kaggle.com/datasets/davilsena/ckdataset>, 2010.
- [4] Yin Cui, Menglin Jia, Tsung-Yi Lin, Yang Song, and Serge Belongie. Class-balanced loss based on effective number of samples, 2019.
- [5] Yin Cui, Menglin Jia, Tsung-Yi Lin, Yang Song, and Serge Belongie. Class-balanced loss based on effective number of samples, 2019.
- [6] Paul Ekman and Wallace V. Friesen. *Manual of the Facial Action Coding System (FACS)*. Consulting Psychologists Press, Palo Alto, CA, USA, 1978.
- [7] Expression in the Wild. Expw dataset. <https://www.kaggle.com/datasets/shahzadabbas/expression-in-the-wild-expw-dataset>.
- [8] FERPlus. Ferplus dataset. <https://www.kaggle.com/datasets/arnabkumarroy02/ferplus>.
- [9] Pierre Foret, Ariel Kleiner, Hossein Mobahi, and Behnam Neyshabur. Sharpness-aware minimization for efficiently improving generalization, 2021.
- [10] Darshan Gera, S. Balasubramanian, and Anwesh Jami. Cern: Compact facial expression recognition net. *Pattern Recognition Letters*, 155:9–18, 2022.
- [11] Priya Goyal, Piotr Dollár, Ross Girshick, Pieter Noordhuis, Lukasz Wesolowski, Aapo Kyrola, Andrew Tulloch, Yangqing Jia, and Kaiming He. Accurate, large minibatch sgd: Training imagenet in 1 hour, 2018.
- [12] Ali Hassani, Steven Walton, Nikhil Shah, Abulikum Abuduweili, Jiachen Li, and Humphrey Shi. Escaping the big data paradigm with compact transformers, 2022.
- [13] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016.
- [14] Chia-Feng Hsu, Sriyani Padmalatha Konara Mudiyansele, Rismia Agustina, and Mei-Feng Lin. Basic emotion detection accuracy using artificial intelligence approaches in facial emotions recognition system: A systematic review. *Applied Soft Computing*, 172:112867, 2025.
- [15] Jie Hu, Li Shen, and Gang Sun. Squeeze-and-excitation networks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018.
- [16] JAFFE. Japanese female facial expression (jaffe) dataset. <https://zenodo.org/records/14974867>.
- [17] Karolinska Directed Emotional Faces. Kdef dataset. <https://kdef.se/download-2/register>, 2002.
- [18] Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. Focal loss for dense object detection, 2017.
- [19] Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. Focal loss for dense object detection, 2018.
- [20] Patrick Lucey, Jeffrey F. Cohn, Takeo Kanade, Jason Saragih, Zara Ambadar, and Iain Matthews. The extended cohn-kanade dataset (ck+): A complete dataset for action unit and emotion-specified expression. In *2010 IEEE Computer Society Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, pages 94–101, 2010.
- [21] Mahmoud Ima. Mma facial expression dataset. <https://www.kaggle.com/datasets/mahmoudima/mma-facial-expression>.
- [22] RAF-DB. Real-world affective faces database. <https://www.kaggle.com/datasets/shuvoalok/raf-db-dataset>.
- [23] Ramprasaath R. Selvaraju, Abhishek Das, Ramakrishna Vedantam, Michael Cogswell, Devi Parikh, and Dhruv Batra. Grad-cam: Why did you say that? visual explanations from deep networks via gradient-based localization. *CoRR*, abs/1610.02391, 2016.
- [24] Ali I. Siam, Naglaa F. Soliman, Abeer D. Algarni, Fathi E. Abd El-Samie, and Ahmed Sedik. Deploying machine learning techniques for human emotion detection. *Computational Intelligence and Neuroscience*, 2022:8032673, 2022.
- [25] Stanford CS231n. Convolutional neural networks for visual recognition (course notes), 2017. Accessed 2026.
- [26] Mukund Sundararajan, Ankur Taly, and Qiqi Yan. Axiomatic attribution for deep networks. *CoRR*, abs/1703.01365, 2017.
- [27] European Data Protection Supervisor. https://www.edps.europa.eu/system/files/2021-05/21-05-26_techdispatch-facial-emotion-recognition_ref_en.pdf, 2021.
- [28] Mingxing Tan and Quoc V. Le. Efficientnetv2: Smaller models and faster training, 2021.
- [29] VCC Tech. Emoset dataset. <https://vcc.tech/EmoSet>.
- [30] Chengpeng Wang, Li Chen, Lili Wang, Zhaofan Li, and Xuebin Lv. Qcs: Feature refining from quadruplet cross similarity for facial expression recognition, 2025.
- [31] Xiang Wei, Ting Feng, Qinghua Huang, Qian Chen, Chao Zuo, and Haigang Ma. Deep learning-powered biomedical photoacoustic imaging. *Neurocomputing*, 573:127207, 2024.
- [32] Sanghyun Woo, Jongchan Park, Joon-Young Lee, and In So Kweon. Cbam: Convolutional block attention module. In

Proceedings of the European Conference on Computer Vision (ECCV), 2018.

- [33] Sanghyun Woo, Shoubhik Debnath, Ronghang Hu, Xinlei Chen, Zhuang Liu, In So Kweon, and Saining Xie. Convnext v2: Co-designing and scaling convnets with masked autoencoders, 2023.
- [34] WSEFEP. Warsaw set of emotional facial expression pictures. <https://www.emotional-face.org/hopsa-sa>.
- [35] Lianghui Zhu, Bencheng Liao, Qian Zhang, Xinlong Wang, Wenyu Liu, and Xinggang Wang. Vision mamba: Efficient visual representation learning with bidirectional state space model, 2024.